

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) now produce high-quality images and videos, yet they can still generate visually plausible samples that violate factual constraints, such as an implausible duplicate object or a hand with the wrong number of fingers. This paper contributes a concrete way to study this broad problem by isolating **counting hallucinations**: generated samples whose object or part counts contradict the counts permitted by the training data. Our first contribution is **CountHalluSet**, a dataset suite with explicit counting criteria across three levels of visual complexity: ToyShape, SimObject, and RealHand. Our second contribution is a standardized evaluation protocol that uses dataset-specific counting models, and a counting-ready indicator for RealHand, to quantify counting hallucination rate separately from other severe generation failures. Our third contribution is a controlled empirical analysis of how solver type, ODE solver order, sampling steps, and initial noise influence counting hallucinations. The results show that common numerical improvements can reduce failures on simple synthetic data but do not reliably reduce counting hallucinations for realistic hands. Finally, we show that widely used perceptual metrics such as Fréchet Inception Distance (FID) do not consistently track counting hallucinations, and we use this finding to motivate a joint-diffusion training paradigm that reduces counting and non-counting failures on RealHand.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become a central generative modeling tool, producing high-quality samples for text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). The contribution problem addressed in this paper begins where this success is incomplete: diffusion models can still fail to preserve factual properties present in the training data. One salient failure is count inconsistency in spatially intricate structures, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Recent work has made important progress in surfacing failure modes of text-guided diffusion models, but much of this progress remains qualitative or prompt-alignment oriented. For example, Borji (2023) summarize qualitative shortcomings in text-to-image models; Liu et al. (2024) use adversarial search to expose undesirable behaviors and failure cases; and Wu et al. (2023) iteratively discover and mitigate spurious correlations. These studies motivate the need for a complementary contribution: a controlled, quantitative setting in which a specific hallucination type can be defined, measured, and compared across sampling conditions.

Hallucination in diffusion models has therefore emerged as a distinct object of study (Aithal et al., 2024). Unlike prompt-alignment errors, which can be probed through CLIP score (Hessel et al., 2021) or VQA score (Huang et al., 2025; Zarei et al., 2024), and unlike obvious degradations such as blur or severe artifacts, hallucinated samples can look plausible while violating factual consistency with the training data. Examples include incorrect object counts, geometries that appear plausible but are physically impossible, and other violations of physical laws. This paper targets the measurement bottleneck created by such subtle failures: current AI systems often do not detect them reliably, and even state-of-the-art large vision-language models such as GPT-4o can struggle with basic finger counting in clear hand images. By introducing a standardized quantification protocol, we aim to

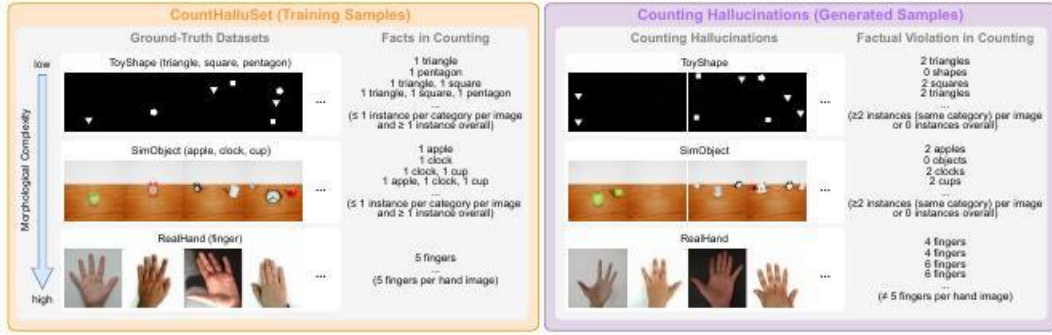


Figure 1: Contribution overview for **CountHalluSet**: training examples with valid counts appear on the left, and model-generated **counting hallucinations** appear on the right. Each dataset makes the counting rule explicit. ToyShape and SimObject allow at most one instance per category and require at least one object overall, whereas RealHand requires exactly five fingers. A generated image is a counting hallucination when it violates these rules, for example by producing two apples or an empty object scene in SimObject, even though such patterns never occur in the dataset.

provide a reviewer-verifiable foundation for studying factual consistency in generative models, an ability that is critical if such models are to serve as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

We focus on counts because they turn a diffuse hallucination phenomenon into a measurable contribution without changing the underlying scientific problem. Among hallucinated attributes such as colors, spatial relationships, geometry, and physical realism, counting errors are comparatively easy to specify and verify. We therefore define **counting hallucination** as follows: *the training data impose predetermined object-count criteria, and a generated image is hallucinated when its counts deviate from those criteria*. For instance, a hand image with six fingers is a counting hallucination under RealHand because such a count never occurs in that dataset, as shown in the last row of Fig. 1. To make this definition operational, we introduce CountHalluSet, which spans increasing morphological complexity across ToyShape (*triangle, square, pentagon*), SimObject (*mug, apple, clock*), and RealHand (*finger*). Figure 1 presents this contribution visually by pairing valid training samples with corresponding counting hallucinations.

Our evaluation protocol contributes the measurement machinery needed to use these datasets systematically. Because counting hallucinations must be separated from generic visual failure, we use counting models tailored to each dataset and, for RealHand, an additional counting-ready indicator. With these tools, we quantify how sampling steps, initial noise, ODE solver order, and solver type affect counting hallucination rates. We also test whether standard perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021), reflect the factual errors captured by counting hallucination rate.

The paper makes the following reviewer-facing contributions:

- **Definition and benchmark.** We define counting hallucinations and introduce CountHalluSet, a suite of ToyShape, SimObject, and RealHand datasets with explicit counting criteria.
- **Evaluation protocol.** We provide a standardized protocol for estimating counting hallucination rate, including dataset-specific counting models and a counting-ready indicator for RealHand.
- **Empirical findings.** We show that common numerical strategies, such as increasing sampling steps from 25 to 100, often fail to mitigate counting hallucinations for human hands even when they help on simpler synthetic datasets.
- **Metric analysis.** We find weak or inconsistent correlations between counting hallucination rate and perceptual metrics, while non-counting failure rate correlates strongly with those metrics.

- **Mitigation direction.** Motivated by the morphology findings, we propose joint-diffusion models and show that they reduce counting hallucinations and non-counting failures on RealHand.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

The hallucination literature motivates our contribution by showing both the importance of the problem and the absence of a quantitative diffusion-specific protocol. Inspired by extensive work on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), recent diffusion-model research has begun to examine hallucinated visual samples. In particular, Aithal et al. (2024) propose a qualitative hypothesis that hallucinated samples may arise from interpolation between distinct data modes. What remains missing, and what this paper targets, is a systematic way to quantify a well-defined hallucination type in diffusion-generated outputs.

Our work is positioned as a measurement contribution rather than a broad taxonomy of all possible hallucinations. By making counting criteria explicit, we can identify hallucinated outputs accurately and study how denoising choices, including sampling steps and ODE solver order, change the severity of the failure. This framing turns the qualitative concern raised by prior work into a controlled empirical question.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The metric literature motivates a second contribution: testing whether common image-generation metrics actually track counting hallucinations. VQA score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are widely used for text-guided image generation, but their reliance on language-based models limits their suitability for detecting subtle hallucinated images. One reason is the statistical lower bound on the tendency of pretrained language models to hallucinate certain fact types, which constrains detection accuracy when language models are used alone (Kalai & Vempala, 2024). Another is the persistent gap between human annotators and LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Perceptual and distributional metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021) are also standard in image generation. Many of these metrics depend on feature embeddings from a pretrained Inception-v3 model (Szegedy et al., 2016), and they are often interpreted as indicators of generative quality. However, *whether they measure factual count consistency is unresolved*. We address this gap by correlating these metrics with counting hallucination rate and non-counting failure rate under controlled diffusion sampling conditions.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

This preliminary section establishes the sampling quantities that our experiments vary and later connect to counting hallucination rates. We focus on the forward noising process, the reverse denoising process, and the ODE-based accelerated samplers because solver type, solver order, step count, and initial noise are the main controlled factors in our contribution.

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $x_0 \sim q_0(x_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define the *forward process* (a.k.a, *diffusion process*) that progressively corrupts data by adding Gaussian noise across T discrete timesteps. This process provides the reference noising trajectory used later when we compare standard Gaussian initialization with diffused initialization. At timestep t , the marginal distribution of the noisy sample x_t conditioned

on the original data $\mathbf{x}_0$ is:

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

Here $t \in \{0, 1, \dots, T\}$, and $\{\beta_t\}_{t=1}^T$ is a predefined time-dependent noise schedule, typically linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021). The factor $\bar{\alpha}_t$ determines the signal-to-noise ratio at step t , moving from $\bar{\alpha}_0 = 1$ with no noise to $\bar{\alpha}_T \approx 0$ with nearly pure noise. Equivalently, Eq. (1) can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

When T is sufficiently large, the terminal marginal $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$ used to initialize sampling. In common DPM implementations (Ho et al., 2020; Nichol & Dhariwal, 2021), T is typically set to 1000, which is the setting our experiments use when reporting ancestral sampling.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) is the generative mechanism whose errors may produce counting hallucinations. Starting from diffused data $\mathbf{x}_T$, DPMs iteratively recover data-like samples by predicting the noise ϵ in $\mathbf{x}_t$ with a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, the *noise prediction model*. The parameters θ are optimized with the objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function. This objective is the common training basis for the models whose counting behavior we evaluate.

Ancestral Sampling. Ancestral sampling is the first solver family in our empirical comparison. Given noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ are obtained by reversing the diffusion process with the same number of steps:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}} (\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the only unknown term and is referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ with the noise prediction model. Following the objective in Eq. (3), the resulting sampling update exposes the role of model prediction error in later discussion:

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. ODE-based accelerated sampling is the second solver family in our empirical comparison and is central to our analysis of sampling steps and solver order. While ancestral sampling in Eq. (4) includes a stochastic term (Kloeden et al., 1992), it has an associated deterministic process that can be written as an ordinary differential equation (ODE), also called the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau) \mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2} \beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2} \beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and starts from ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. We set the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. Applying the variation-of-constants formula gives the exact one-step solver:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ lets ODE-based solvers accelerate sampling with fewer, larger steps than ancestral

sampling; the total number of sampling steps is $T/\Delta t$. Because the linear part $\mathcal{G}(t_2, t_1)$ is available in closed form, accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) mainly allocate numerical effort to approximating the nonlinear integral involving h . This is precisely why solver order and step count are natural experimental factors for our counting-hallucination study. Additional theoretical context appears in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section makes the paper’s measurement contribution explicit. We first formalize counting criteria and define counting hallucinations as violations of those criteria. We then instantiate the definition in CountHalluSet, specifying the counting rules, counting models, and evaluation protocol for ToyShape, SimObject, and RealHand.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We define counting hallucinations in a way that is dataset-grounded and directly measurable. The key idea is to treat the training dataset as imposing explicit counting facts, then mark generated samples as hallucinated only when they are countable and violate those facts.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ be a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the permitted number of objects in category c is encoded by a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$, such as $\{0, 1\}$ or $\{1, 2, 3\}$, so that $N_c(\mathbf{x}) \in \mathcal{S}_c$. We also require every valid reference sample to contain at least one object, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of categories under consideration. These rules are the count-level facts that our evaluation asks generated samples to respect.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we identify a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ only when both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must have enough visual quality for the background and any present objects to be clearly discernible. This condition separates the paper’s focal contribution, count-based factual error, from generic severe degradations. We define a binary *counting-ready indicator* $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, where 1 denotes sufficient quality for counting and 0 denotes otherwise. In practice, the indicator may be obtained from human annotation, from a classifier calibrated to human annotation, or from other automated criteria.
2. **Violation of counting facts.** A generated image violates the counting facts when some category c has $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the image contains no objects, $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$, despite the dataset requiring at least one object. We operationalize this condition with a counting model that predicts object counts in generated images.

A counting hallucination (CH) is therefore a countable sample whose predicted counts deviate from the dataset criteria: $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$ and either $\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$. Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

CountHalluSet is the benchmark contribution that instantiates the definition across increasing visual complexity. It consists of ToyShape, SimObject, and RealHand, each with explicit count rules and a matching evaluation procedure. Examples from the datasets and their corresponding counting hallucinations appear in Fig. A1 and A2, with additional statistics in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. ToyShape is the simplest CountHalluSet component and serves as a controlled setting where countable objects have minimal morphology. It contains 30,000 images with three

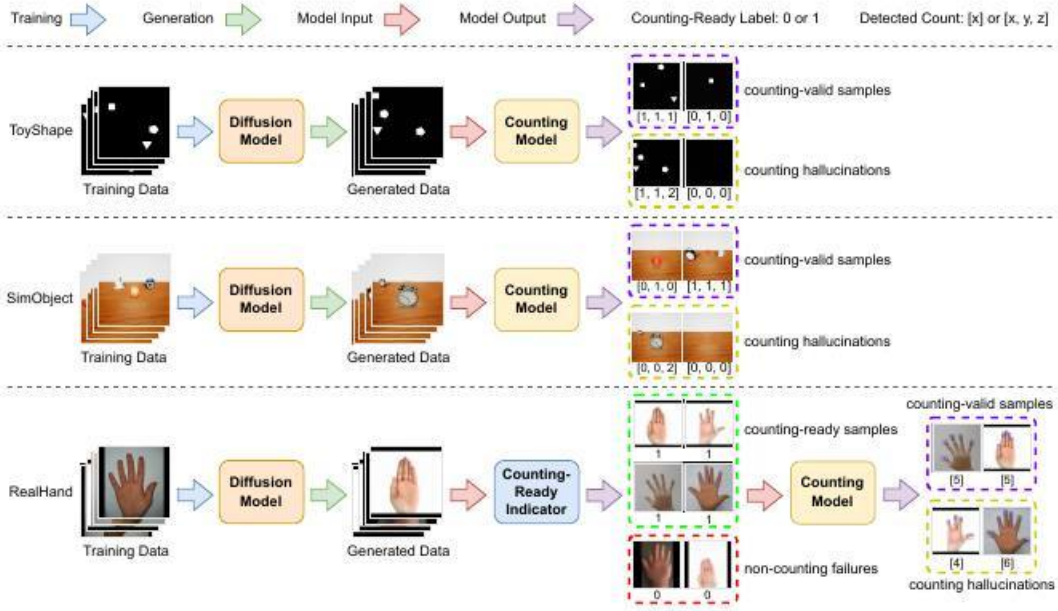


Figure 2: Standardized evaluation protocol for the CountHalluSet contribution. ToyShape and SimObject samples are directly passed to dataset-specific counting models. RealHand adds a counting-ready indicator, implemented as a binary classifier aligned with human judgments, to separate counting hallucinations from non-counting failures such as severely deformed fingers. Count labels are ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; RealHand: *finger*.

geometric shapes: *triangle, square, and pentagon*. All shapes are white, have an equal area of 120 pixels, and never overlap within an image. The counting rule is explicit: each image contains at most one instance of each shape category and at least one shape overall, as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle, square, pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, a ToyShape generation is a counting hallucination if it contains two or more shapes from the same category, such as two pentagons in the top-right region of Fig. 1. Empty images containing only the black background are also counting hallucinations because the dataset always contains at least one shape. Additional examples are provided in Appendix A.5.

Counting model used in ToyShape. For ToyShape, the protocol uses a counting model directly and does not require a counting-ready indicator, because severe visual degradation is rare in this simple domain. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape samples in which each category appears 0–3 times per image (i.e., $S_c = \{0, 1, 2, 3\}$). This expanded training range covers both valid and hallucinated count patterns, enabling accurate quantification. The resulting model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For each diffusion model trained on ToyShape, we generate 30,000 samples, matching the training-set size. We then predict object counts for every generated image and mark samples that violate the ToyShape criteria as counting hallucinations. The standardized evaluation path is summarized in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. SimObject is the intermediate CountHalluSet component, preserving simple count rules while increasing visual realism and object morphology. It contains 30,000 rendered images of everyday objects across three categories: *mug, apple, and clock*, each with 10 intra-class variations. Objects are randomly placed on a wooden tabletop under fixed lighting, with little or no mutual occlusion, as shown in the middle-left of Fig. 1. The dataset is generated in Unreal Engine 5 (Epic Games, 2024) with photorealistic rendering of object geometry, materials, lighting, and reflections. Its counting criteria match ToyShape (see Sec. 4.2.1). Formally, we have $C = \{\text{mug, apple, clock}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject shares ToyShape’s count structure, its counting hallucination definition, counting-model construction, and quantification protocol follow the same procedure, as shown in Fig. 2. This design isolates the effect of increased visual and morphological complexity while keeping the counting rule fixed.

4.2.3 REALHAND

Dataset description. RealHand is the most realistic CountHalluSet component and tests whether the protocol remains useful for morphologically complex objects. It contains 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate raw images from these sources and remove images with blur or artifacts. Each remaining image depicts one human hand with five fingers, covering dorsal (back) and palmar (front) views across varying lighting conditions and backgrounds, such as white surfaces, blackboards, and walls, as shown in the lower-left of Fig. 1. The counting criterion is exact: $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. RealHand requires an additional distinction that is central to our evaluation contribution. Generated hand images may fail because of incorrect finger counts, but they may also fail because of blurriness, severe deformation, or visual artifacts that make counting unreliable. We therefore use a counting-ready indicator, implemented as a binary classifier, to separate countable samples from non-counting failures. Examples of both counting hallucinations and non-counting failures appear in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To implement this separation, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as the counting-ready indicator aligned with human judgments. For finger counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k generated images with fingertip annotations. On 500 randomly sampled images evaluated by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Appendix A.6 provides examples of the YOLO detections used by the protocol.

Quantification of counting hallucinations. The RealHand quantification protocol therefore differs from ToyShape and SimObject in one reviewer-relevant step. For ToyShape and SimObject, all generated images are directly counted because severe failure cases are rare. For RealHand, we first apply the counting-ready indicator to retain hand images with sufficient visual quality, and then apply the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. The experimental setup is designed to make counting-hallucination comparisons controlled across the three CountHalluSet datasets. We train a DDPM (Ho et al., 2020) from scratch on ToyShape and fine-tune latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on CelebA-HQ (Karras et al., 2018), on SimObject and RealHand. For all models, the timestep count T is 1,000 with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and 256×256 for LDMs. We use Adam (Kingma, 2015) with learning rate 0.0001, train for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively, and use an effective batch size of 256.

Inference of Diffusion Models. At inference time, we vary the sampler choices that our contribution seeks to analyze. For ancestral sampling (i.e., DDPM), we use 1,000 sampling steps. For ODE-based sampling, we compare a first-order method, DPM-Solver-1, equivalent to DDIM (Song et al., 2021), with a second-order method, DPM-Solver-2 (Lu et al., 2022). Each ODE solver is evaluated at 25, 50, and 100 denoising steps. For RealHand, the YOLO counting model uses a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless stated otherwise, each evaluation generates the same number of samples as the corresponding training set, and all reported results are averaged over three random seeds.

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

Table 1: Main quantification results for the CountHalluSet contribution. Counting hallucination rate (CHR; counting hallucinations divided by total generated samples) is reported under solver configurations and initial-noise settings for ToyShape, SimObject, and RealHand. For RealHand, non-counting failure rate (NCFR) reports failures detected by the counting-ready indicator, and total failure rate (TFR) is CHR plus NCFR. *Diffused* and *Normal* denote ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 presents the central empirical contribution: counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under controlled changes to solver type, ODE solver order, sampling steps, and initial noise across ToyShape, SimObject, and RealHand. The results support five reviewer-facing observations.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but exacerbates them in RealHand.** As shown in Table 1, increasing ODE sampling steps from 25 to 50 to 100 generally lowers CHR on ToyShape and SimObject, but tends to raise CHR on RealHand, except for the 100-step DPM-Solver-2 case. This contrast is important for the paper’s contribution: a numerical strategy that improves count consistency on simple synthetic distributions does not transfer reliably to realistic hands. The pattern suggests that finer solver granularity benefits simpler, more structured datasets, whereas additional steps on RealHand may amplify local inconsistencies in a more complex distribution.

(2) **Employing higher-order ODE solvers effectively mitigates non-counting failures but exacerbates counting hallucinations in RealHand under the standard inference steps.** As shown in Table 1, comparing DPM-Solver-1 with DPM-Solver-2 on RealHand at standard step counts such as 25 and 50 reveals a clear trade-off. The higher-order solver substantially reduces NCFR, but it also produces higher CHR. Increasing the number of sampling steps shows a related pattern for both DPM-Solver-1 and DPM-Solver-2. Thus, “better” numerical sampling conditions can improve perceptual usability, reflected in fewer non-counting failures, without enforcing the factual correctness of finger counts.

(3) **DDPM substantially outperforms ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across the reported comparisons, ancestral sampling (i.e., DDPM) achieves the lowest CHR, NCFR, and TFR. This observation contributes a practical reference point: DDPM can serve as a strong lower-failure baseline when quantifying hallucinations under the same trained model and dataset. It also suggests that ancestral sampling is an effective strategy for minimizing the failures measured by our protocol.

(4) **A better initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise, i.e., ground-truth initial noise, generally yields lower CHR, NCFR, and TFR than “normal” standard Gaussian noise. This observation is a supported contribution about initialization: a more informed initial noise distribution can reduce both counting hallucinations and non-counting failures. The likely explanation is that diffused noise better matches the initial noise distribution encountered during training.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that diffusion models tend to produce higher CHR as countable objects become morphologically more complex. This cross-dataset pattern is the empirical bridge from the benchmark contribution to the mitigation contribution in Sec. 5.5. It suggests that fine-grained details and complex spatial relationships make

Table 2: Metric-audit results comparing counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) against FID and MUSIQ. PTI-Hand results use a pretrained Stable-Diffusion-3.5-Medium model prompted with ten hand-generation prompts to produce 5050 samples, matching RealHand. PTI-Hand FID is computed between RealHand images and generated samples. Here, p is the p-value for the significance of r or ρ . † indicates DDPM results integrated with ODE-solver results across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand	RealHand	PTI-Hand
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01	-0.97	<0.01	-0.99	<0.01	-0.97	<0.01
Pearson† (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—	-0.98	<0.01	—	—	-0.98	<0.01
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29	-0.88	<0.01	-0.46	0.29	-0.88	<0.01
Spearman† (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—	-0.91	<0.01	—	—	-0.91	<0.01
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—	0.97	—	0.99	—	0.97	—
Distance Correlation† (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—	0.97	—	—	—	0.97	—

it harder for diffusion models to preserve accurate object counts, resulting in more frequent additions or omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

This subsection contributes a metric audit: it tests whether standard perceptual metrics measure the factual count errors introduced above. Because counting hallucination is a form of factual hallucination, a natural question is: *Does a better perceptual metric, such as lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our answer is **not necessarily**.

To examine this relationship, we correlate CHR with perceptual metrics and compare the result with correlations between NCFR and the same metrics. We use two common perceptual measures: the distribution-based FID (Heusel et al., 2017) and the instance-based MUSIQ (Ke et al., 2021). We estimate correlation with three complementary statistics: Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The evidence below explains why perceptual quality and counting factuality should be reported separately.

✦ **The correlations between CHR and perceptual metrics are dataset-dependent and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, Pearson correlations between CHR and FID change substantially across datasets: positive for SimObject, negative for RealHand, and positive again for PTI-Hand. The corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also drop from 0.82 for SimObject to 0.74 for RealHand and 0.47 for PTI-Hand, indicating that more complex generated content weakens the association between CHR and FID. When DDPM results are integrated with ODE-solver results, these correlations weaken further.

✦ **The correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast, NCFR is strongly and robustly associated with perceptual metrics across settings. For NCFR vs. FID, all reported correlation coefficients exceed 0.88 and have very small p-values, indicating high statistical significance. This contrast supports the paper’s metric contribution: perceptual metrics capture broad visual failure more reliably than they capture count-level factual hallucination.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

This subsection tests whether the RealHand findings are an artifact of our controlled training setting or transfer to a large pretrained text-to-image model. We evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) with the same counting-ready indicator and counting model introduced in Sec. 4.2.3; additional details appear in Appendix A.8. As shown in Figure 3, the effects of solver type and sampling steps on counting hallucinations in the pretrained model align closely with the trends from the controlled RealHand experiments. This consistency strengthens the validity of the findings in Sec. 5.2 and supports the broader applicability of the evaluation protocol. The correlation results in Table 2 provide additional support: CHR remains weakly related to perceptual metrics, whereas

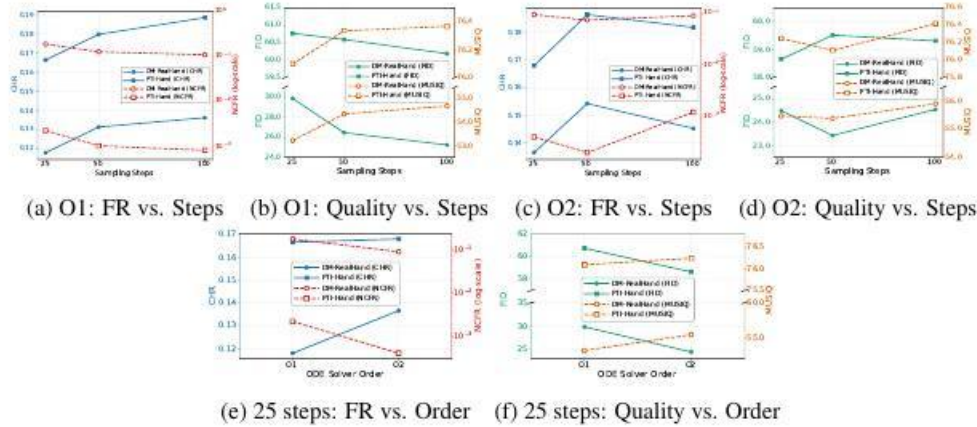


Figure 3: Generalization evidence comparing trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers: DPM-solver-1 / DPM-solver-2 for DM-RealHand and Euler / Heun for PTI-Hand. FR metrics include CHR and NCFR, while quality metrics include FID and MUSIQ. Across sampling-step and solver-order variations, both models show highly consistent trends.

NCFR remains strongly related to them. Together, these results show that the proposed protocol and conclusions transfer beyond models trained only in our controlled setting.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

This subsection turns the empirical findings into a mitigation direction while preserving the scope of the paper’s evidence. Observation 5 in Sec. 5.2 shows that simpler countable objects have lower counting hallucination rates, suggesting that diffusion models preserve count-based factuality more reliably when the relevant structure has low morphological complexity, such as in masks. This motivates a concrete question: can a diffusion model reduce counting hallucinations by denoising RGB visual representations together with explicit structural constraints, such as object masks that encode basic topology and counts, in a shared latent space? We call this joint learning paradigm **joint-diffusion models (JDM)**.

Concretely, JDM concatenates the latent representation of a hand mask channel-wise with the latent representation of the corresponding RGB image, as shown in Fig. 4. We fine-tune the UNet pretrained on CelebA-HQ (Karras et al., 2018), following the RealHand configuration in Sec. 5.1, using RealHand images and associated masks extracted by SAM-2 (Ravi et al., 2024). To support both modalities, we expand the denoising UNet input and output channels to match the combined dimensionality of the image and mask latents, i.e., $3+3=6$ channels.

As shown in Table 3, JDM consistently lowers both CHR and NCFR relative to the vanilla LDM across sampling conditions, supporting it as a promising mitigation direction for structurally intricate objects with counting ambiguity. The trade-off is that JDM yields higher FID. This degradation likely reflects the additional denoising burden of jointly modeling two modalities in a small-scale training regime, which shifts capacity toward structural realism at the expense of perceptual quality. We also observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Appendix A.9 provides extended details, results, and discussion.

6 CONCLUSION

This paper contributes a systematic framework for making diffusion-model hallucinations measurable through the focused case of counting hallucinations. We introduce **CountHalluSet**, define dataset-specific counting criteria, and provide a practical protocol that separates counting hallucinations from non-counting failures. Using this framework, we show that common numerical strategies, including more sampling steps and higher-order ODE solvers, do not reliably reduce counting hallucinations

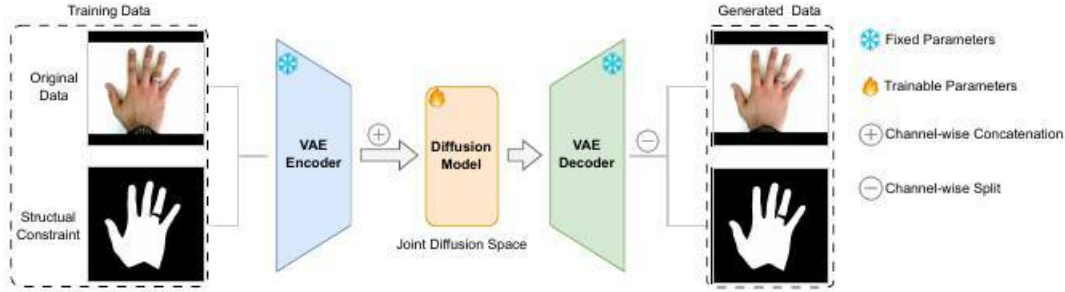


Figure 4: Mitigation contribution: the proposed joint-diffusion model (JDM). JDM concatenates latent representations of hand RGB images and corresponding segmentation masks channel-wise. Joint denoising encourages anatomically plausible hand poses and correct finger topology during generation, reducing CHR and NCFR.

Table 3: Mitigation results comparing vanilla LDM with the proposed JDM on RealHand across ODE solvers and sampling steps. Lower failure rates indicate better counting and visual validity. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs for comparability with the LDM baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

for realistic hands, even when they improve simpler datasets or reduce non-counting failures. We further show that perceptual metrics such as FID and MUSIQ do not consistently reflect count-level factual errors, while they correlate strongly with non-counting failures. Experiments on pretrained text-to-image generation support the generality of these findings, and the joint-diffusion results provide an initial mitigation direction by coupling RGB generation with structural mask constraints.

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A APPENDIX

A.1 PROOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

This appendix provides the derivation supporting the sampling-error framing used in the main text. For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the learned reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence is:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Using the Markov property, both reverse processes decompose into their terminal distributions and stepwise transition kernels:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule then decomposes the trajectory-level KL divergence into an initial-distribution term and accumulated transition terms:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalizing variables cannot increase KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) and Eq. (A.5) yields the bound on total accumulated distributional error:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Equality would hold only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, meaning the diffusion model exactly recovers the data distribution $q(\mathbf{x}_0)$. This upper bound on the overall KL divergence $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ between real and generated data matches the DDPM variational bound (Ho et al., 2020) after excluding data entropy $H(\mathbf{x}_0)$. Thus, the derivation clarifies why DDPM’s variational loss is relevant to the distributional discrepancy that underlies our sampling-condition analysis.

Diffused-prior gap. The first error source is the mismatch between the ground-truth diffused distribution $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the prior used during sampling, $p_\theta(\mathbf{x}_T)$. This mismatch appears because the number of diffusion steps T is finite in practice. It can be quantified with a distributional metric such as:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes Kullback–Leibler (KL) divergence. Other metrics, including FID or Wasserstein distance, can also be used to evaluate this discrepancy. Because the diffused-prior gap is present at the start of sampling, it can propagate through denoising and contribute to persistent output error.

Noise model prediction error. The second error source is the noise prediction model. The estimate $\epsilon_\theta(\mathbf{x}_t, t)$ approximates, but need not exactly equal, the true noise ϵ , even though such approximation is supported in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. From Eq. (5), any error in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

A.2 PROOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver approximates the solution as example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, $\hat{\mathbf{x}}_k$ represent the numerical trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$ and $\hat{h}$ is the estimated model function. For a specific numerical trajectory from $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$, we want to calculate its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence repeatedly:

$$\begin{aligned} e_N &= \mathcal{G}(\tau_N, \tau_0) e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \\ &\quad \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \end{aligned} \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term represents the deterministic impact on the final path error of a specific initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

This appendix clarifies a technical distinction behind the sampler comparisons in the main experiments. As shown in Eq. (2), the DDPM (Ho et al., 2020) forward process is:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

Its reverse process is:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16), but the deterministic reverse process with $\eta = 0$ is:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Thus, although DDPM and DDIM share the same forward diffusion process, DDIM is not obtained by simply removing the stochastic term from the DDPM reverse process. The two methods use different coefficient formulations for the predicted noise term, which helps motivate reporting sampler-specific counting hallucination results.

A.4 DATASET EXAMPLES AND DETAILS

This appendix supports the CountHalluSet contribution with examples and count statistics. Figure A1 shows representative samples from ToyShape, SimObject, and RealHand. As shown in Table A1, each dataset is balanced across its categories, and Table A2 shows comparable distributions of image-level object counts. These summaries help rule out simple category-count or object-count imbalance as the driver of the hallucination trends reported in the main text. The datasets appear in the same order in both tables, and the category order in Table A1 is:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

This appendix illustrates the failure type quantified by CHR. Fig. A2 shows representative counting hallucinations for all three datasets. For ToyShape and SimObject, hallucinations include samples with more than one instance of at least one category or samples with no objects despite a normal

Table A1: Category-level object counts for each CountHalluSet dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Image-level object-count distribution for each CountHalluSet dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting hallucinations are generated hands with incorrect valid-finger counts, such as 4 or 6 fingers, as shown in Fig. A2(c). The last two images of the first row in Fig. A2(c) may appear to contain five fingers, but one finger is partially incomplete and visibly artifacted, leaving only four valid fingers; these samples are therefore counted as hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

This appendix documents the fingertip detector used in the RealHand protocol. Figure A3 presents representative YOLO predictions on generated RealHand images. The detector accurately identifies fingertips across diverse hand poses and orientations, including dorsal and palmar views. However, because the protocol measures finger count rather than full anatomical correctness, YOLO may still treat anatomically implausible but five-fingertip configurations as counting-valid. Examples include duplicated thumbs, such as the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a), or five middle fingers without a thumb, such as the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a).

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts incorrect fingertip counts, such as 3, 4, 6, or 7 fingertips. Figure A3(c) shows rare edge cases (<1%) where unclear boundaries, cracked fingers, or floating artifacts prevent reliable detection. These rare cases can be partly mitigated by tuning YOLO inference hyperparameters, such as IoU and confidence thresholds, but doing so may reduce performance on the majority of counting-valid samples. Because we use one consistent detection setup for every RealHand evaluation, these edge cases do not change the comparative conclusions, although they remain a limitation of automated counting.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

This appendix visualizes the three-way RealHand taxonomy that separates CHR from NCFR. Figure A4 shows counting-valid samples, counting-hallucinated samples, and non-counting failures. As illustrated in Fig. 2, the RealHand protocol first applies the counting-ready indicator to separate counting-ready samples, including both counting-valid and counting-hallucinated images, from images unsuitable for counting. Unsuitable images typically contain severe artifacts such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then classified as valid or hallucinated depending on whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

For the PTI-Hand generalization experiment, we use 10 hand-generation prompts with Stable-Diffusion-3.5-Medium. The prompts are:

1. A single human hand, front view, all fingers in frame and visible.

2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of counting hallucination assumes that the model generates a count pattern absent from the training data. For large pretrained diffusion models, the full training distribution is inaccessible, so this assumption cannot be verified directly. To keep the PTI-Hand evaluation conservative and reproducible, we label generated hand images with six or more fingers as counting hallucinations. This criterion is justified because real images may show fewer than five visible fingers because of occlusion, articulation, or viewpoint, but images of hands with six or more fingers are highly unlikely in real data. Thus, finger counts ≥ 6 provide a reliable empirical signal for counting hallucination in the pretrained-model setting; examples appear in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM results in Table 3, we fine-tune the denoising UNet for 120k steps, compared with 80k steps for the vanilla LDM. This larger training budget is needed because extending the latent space from three to six channels creates a substantial mismatch with the pretrained LDM latent distribution. Additional optimization helps close this distributional gap and lets the JDM modeling capacity emerge.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Extended mitigation results on RealHand comparing fine-tuned SD-1.5 with SD-1.5 adapted through the joint-diffusion training paradigm. The first row is the baseline obtained by fine-tuning Stable-Diffusion-1.5 on RealHand. The second row freezes all intermediate layers of the baseline and fine-tunes only the input and output layers with extended channel dimensions. Lower values indicate better performance.

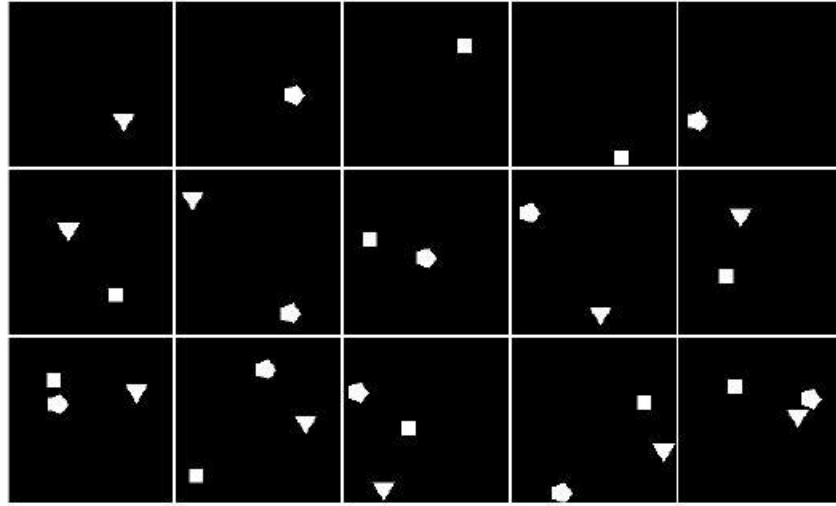
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

JDM should be interpreted as a **training paradigm** for improving factual generation rather than as a superior inference strategy. It extends only the input and output channels of the UNet while keeping intermediate layer sizes unchanged, so it introduces negligible inference-time overhead relative to a vanilla LDM. At the same time, adding mask channels to a pretrained text-to-image model disrupts the manifold geometry of the pretrained latent space and changes the original denoising distribution, making direct fine-tuning nontrivial.

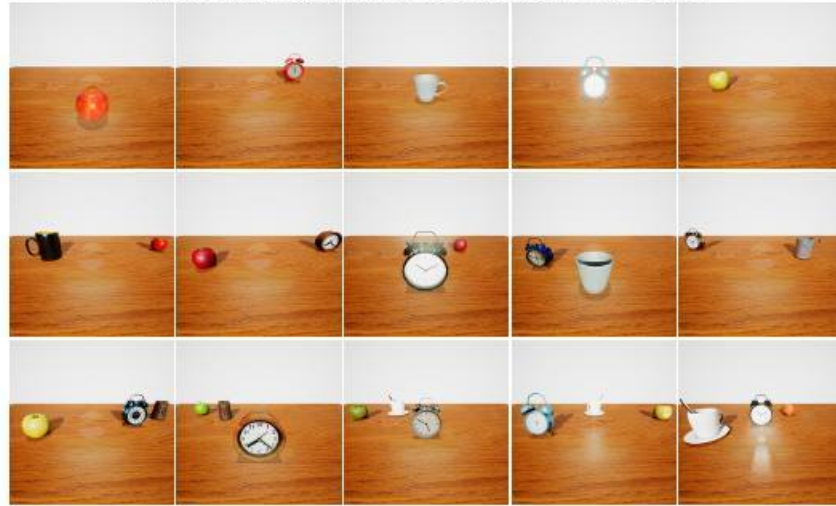
Despite this challenge, JDM still produces promising reductions in counting hallucinations for large pretrained diffusion models under controlled fine-tuning. In practice, aligning the extended JDM

latent space with the rigid latent geometry of large pretrained models is difficult at the limited scale of RealHand. We therefore use a two-stage fine-tuning strategy: first fine-tune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and fine-tune only the input and output layers with extended channels. As shown in Table A3, this constrained setup still lowers counting hallucination rates at 25 and 50 sampling steps across both solvers.

Overall, the appendix reinforces the main-paper claim that JDM is a promising direction for enforcing factual consistency in image generation. Exploring its behavior under true large-scale training remains an important direction for future work.



(a) ToyShape examples with valid geometric-shape counts

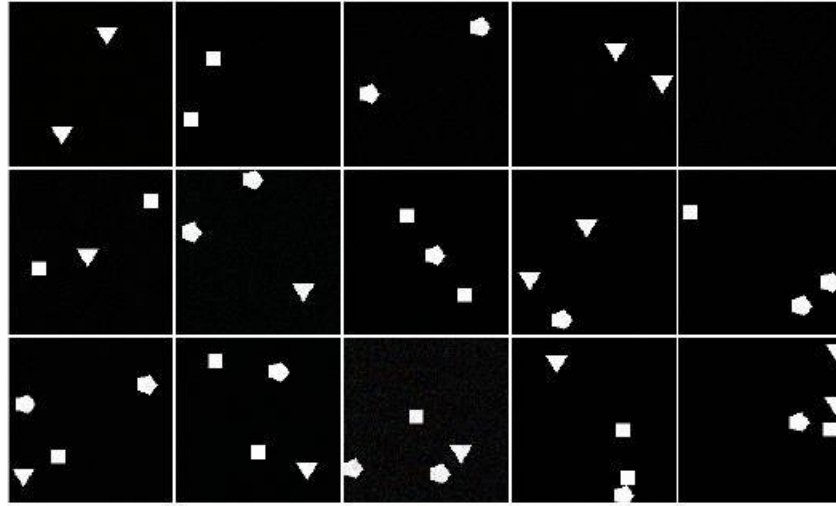


(b) SimObject examples with valid everyday-object counts



(c) RealHand examples with valid five-finger counts

Figure A1: Representative CountHalluSet examples used to define valid count criteria for ToyShape, SimObject, and RealHand.



(a) ToyShape counting hallucinations that violate the dataset's shape-count rule

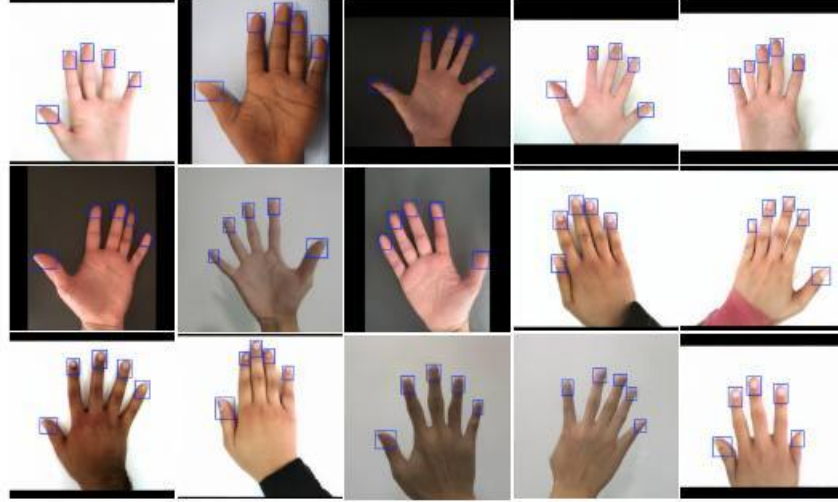


(b) SimObject counting hallucinations that violate the dataset's object-count rule

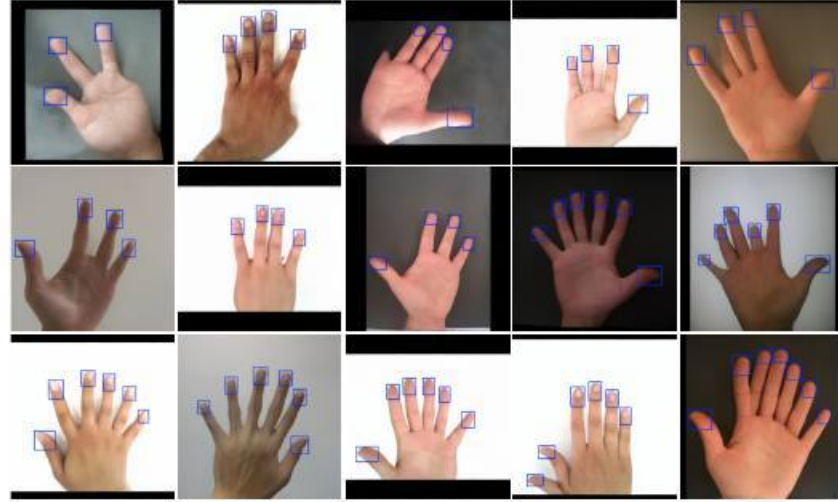


(c) RealHand counting hallucinations with invalid finger counts

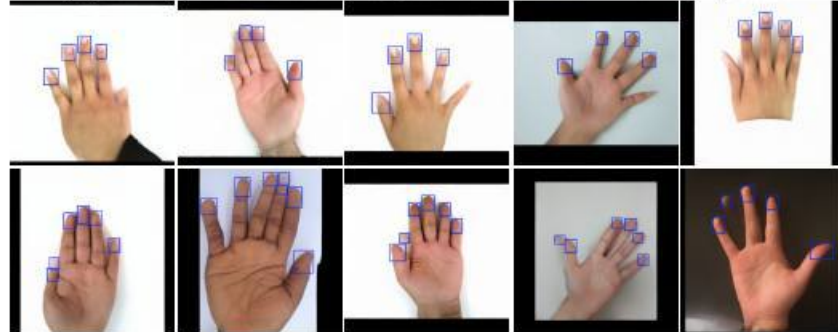
Figure A2: Representative counting hallucinations used to illustrate CHR across ToyShape, SimObject, and RealHand.



(a) Counting-valid YOLO predictions with five detected fingers

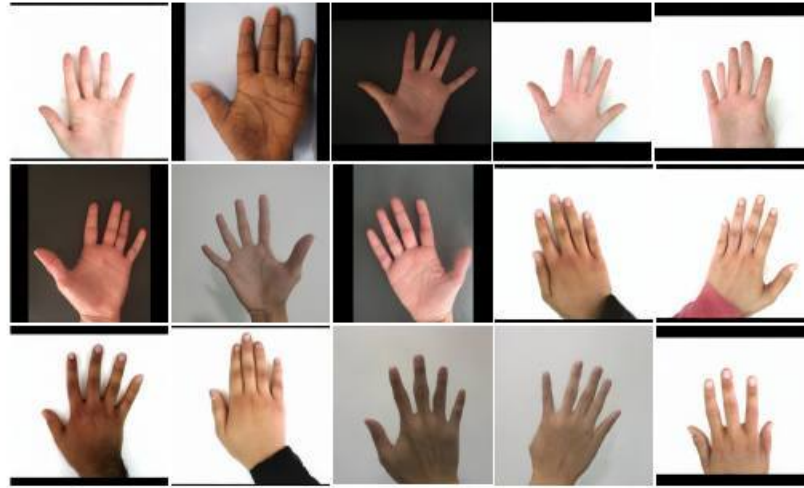


(b) Counting-hallucinated YOLO predictions with a non-five fingertip count

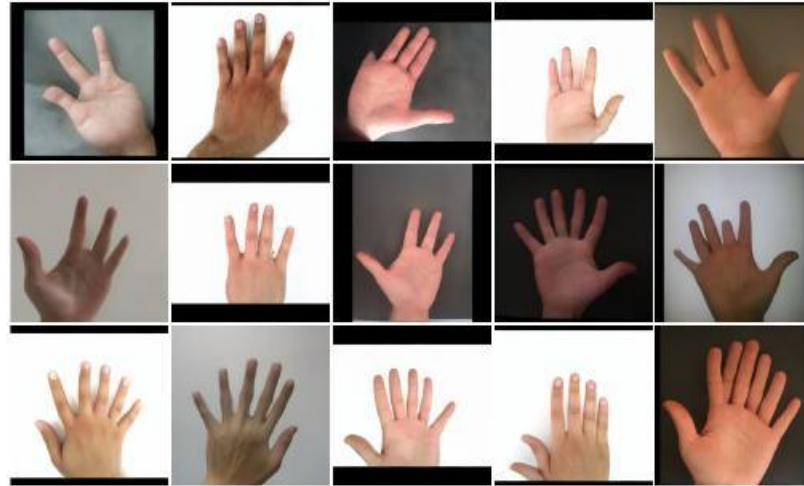


(c) Rare edge cases where YOLO predictions are unreliable

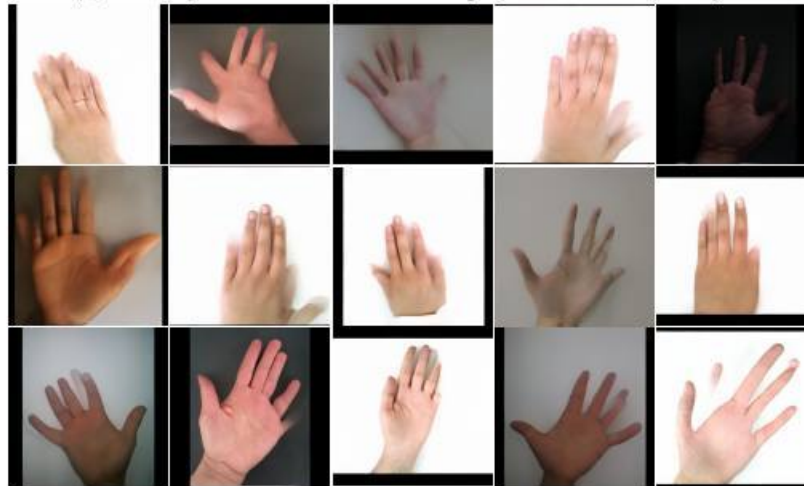
Figure A3: YOLO prediction examples supporting the RealHand counting protocol. (a) Counting-valid samples with five fingertip boxes predicted by YOLO, including cases with two thumbs. (b) Counting-hallucinated samples with incorrect fingertip counts, such as 3, 4, 6, or 7 detections. (c) Rare edge cases where the counting-ready indicator does not filter the sample and YOLO struggles with blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid RealHand samples with five valid fingers



(b) Counting-hallucinated RealHand samples without five valid fingers



(c) RealHand non-counting failures unsuitable for counting

Figure A4: Three RealHand outcome classes used to separate CHR from NCFR. (a) Counting-valid images have five clearly discernible fingers. (b) Counting-hallucinated images have an incorrect number of valid fingers, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe artifacts such as distorted, cracked, partially visible, or floating fingers.



(a) PTI-Hand counting-valid samples with five valid fingers



(b) PTI-Hand counting hallucinations with more than five fingers



(c) PTI-Hand non-counting failures unsuitable for counting

Figure A5: Three generated-hand outcome classes for pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid images have five clearly discernible fingers. (b) Counting-hallucinated images have more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe artifacts such as distorted, cracked, partially visible, or floating fingers.